Lab 1 – AI² Product Description

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1 Introduction

Many organizations handle tens of thousands of invoices annually, often through manual processes that create inefficiencies and risk. According to the AI Invoice Processing Benchmarks for 2025, manually processed invoices can cost anywhere between $12-$14 each, as opposed to the reduced cost of just $3-$6 per invoice processed when using automated systems. The reason for these high costs comes from manual processes often involving repetitive data entry and going through numerous verification steps. As a result, there are slower turnaround times, an increased potential for human error, and mounting backlogs that put a strain on an organization's operations. In short, manual processing is expensive, slow, and prone to errors.

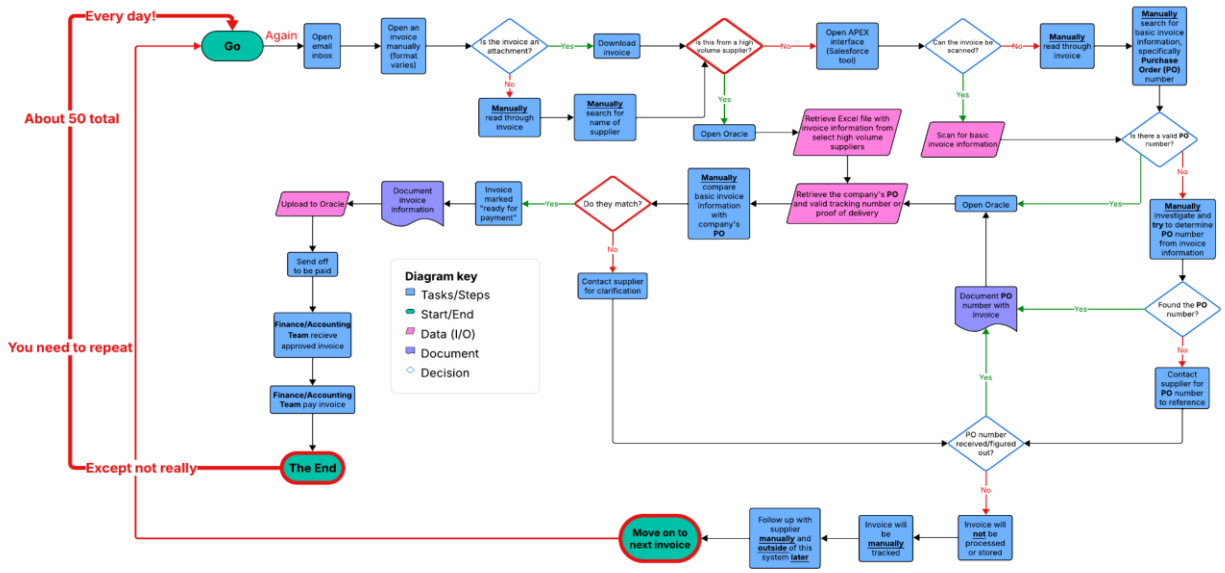
Atlantic Diving Supply, also known as ADS, currently struggles with this same issue. Their Accounts Payable, or AP, team is facing a high-volume of invoices that they must manually process every day. According to ADS, the AP team receives approximately 150,000 invoices annually from about 3,000 suppliers. However, their AP team only has six AP clerks on staff where each member is responsible for processing around 50 to 100 invoices a day, every day. A simple calculation shows that the AP team as a whole needs to process roughly 300 to 600 invoices daily. This volume creates significant pressure, especially when considering that each invoice must be manually reviewed and matched against corresponding Purchase Order (PO) numbers.

Unfortunately, according to ADS, this manual process has created extra work on their other departments and any errors made had to be manually tracked. One prominent error has been over receipts in which the AP team is physically receiving items that have not been shipped in full and there is no way to remove this mistake when it occurs. One such example as provided by ADS was

if a purchase order for 10 of an item was placed but a supplier only invoices and ships 1 of the item, yet the AP representative processes it as if it were for the entire 10 items originally ordered. Additionally, the delays caused from manual processing have raised concerns from management in ADS regarding timely processing and how their current manual processing is not efficient. The AP staff spend long hours performing low-value verification tasks rather than focusing on analyzing tasks which are of higher value. The current process flow, as demonstrated in Figure 1, highlights these inefficiencies embedded in this system through the lens of the daily struggle an ADS AP clerk goes through to process invoices.

Figure 1

Current Process Flow



The impact of this manual invoice processing system is felt across multiple stakeholders. First, the ADS Accounts Payable team faces daily stress and have a limited capacity for higher-level financial tasks, suffering the brunt of the operational burden from needing to manually process invoices day-to-day. Secondly, suppliers awaiting payment experience delays and inconsistent

communication, which can diminish trust and disrupt supply chain continuity. Lastly, there is the management team that rely on timely and accurate financial data to monitor performance, forecast cash flow, and make strategic decisions. These ripple effects of inefficiency in invoice handling extend beyond just the AP department but across multiple ones, influencing the broader financial health and operational agility of the ADS company.

To address these challenges, this paper introduces AI², otherwise known as Artificially Intelligent Invoices, as a web-based intelligent automation system designed to transform invoice management. AI² aims to reduce the manual workload of the AP team, accelerate invoice processing, and improve data accuracy through AI-driven recommendations and adaptive learning. The system combines automated parsing and matching of invoice and PO data with confidence scoring to assess the reliability of each match. Additionally, AI² incorporates a human-in-the-loop learning to ensure that feedback from AP clerks is taken in to continuously refine the system's performance. By integrating automation with human oversight, AI² enhances efficiency while maintaining control and transparency. With this approach, the AP team can feel empowered to shift from reactive verification to proactive financial management, improving outcomes with suppliers, staff, and leadership.

2 Product Description

2.1 Overview

AI² is a web-based automation system that streamlines invoice intake, organization, and purchase order matching. Although designed initially for ADS, the system is scalable and adaptable for any organization with similar high-volume invoice processing needs across diverse

formats. Its primary objective is to accelerate payment readiness and reduce manual workload without sacrificing accuracy, transparency, or human control.

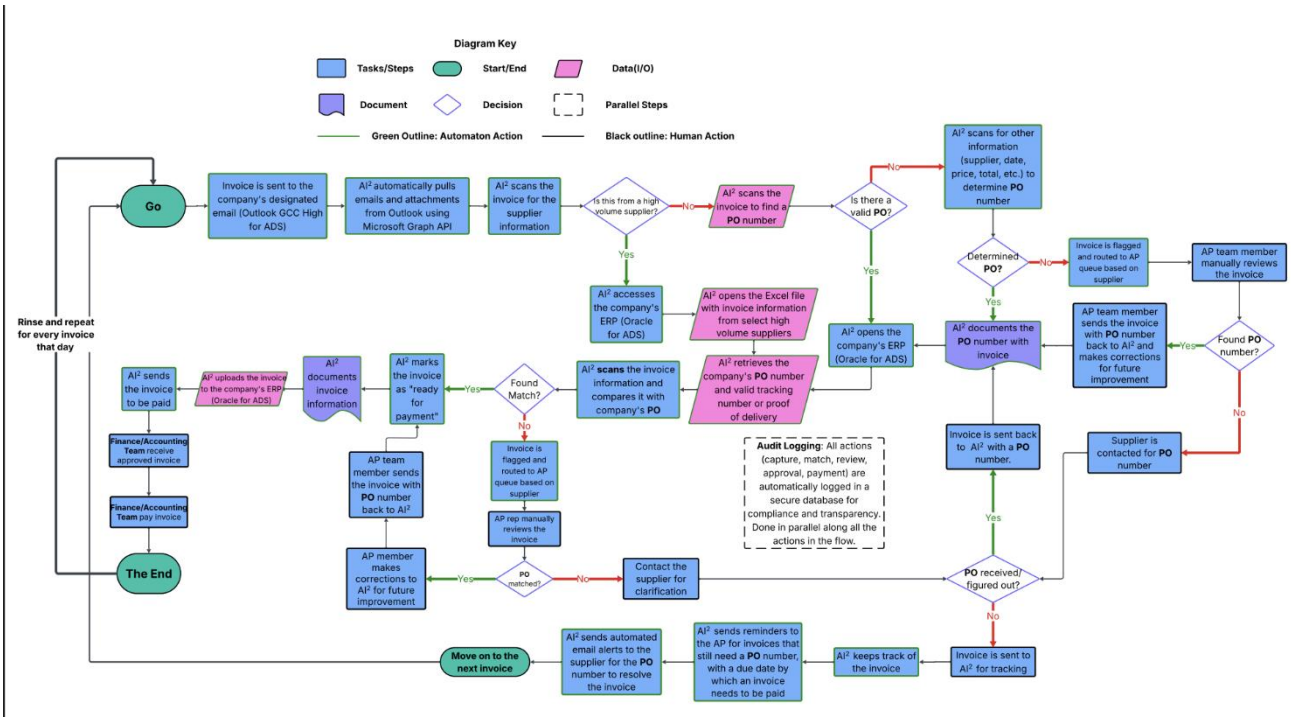
2.2 Key Product Features and Capabilities

Some of the key features include how the system is integrated with Outlook GCC High, its smart matching engine, confidence scoring, human-in-the-loop learning, supplier-based queue assignment, and audit logging for transparency. AI² automates the entire invoice intake process by integrating with Outlook GCC High using the Microsoft Graph API, allowing it to automatically retrieve and parse invoice attachments and email body text in search of invoice details. Its smart matching engine connects to Oracle NetSuite through Alteryx API to retrieve purchase order data and match invoices to the correct PO, line items, and amounts.

Each match is then evaluated using a confidence scoring algorithm where high-confidence cases are automatically processed and low-confidence cases are flagged for human review to be sent into queues. These queues are supplier-based in that the system routes invoices to the appropriate AP representative automatically based on supplier name while allowing reassignment by administrators to be done at the supplier level for each AP representative. During human corrections, the system incorporates human-in-the-loop learning by storing these corrections made by the AP staff on the flagged invoices as training data, enabling the model to improve accuracy over time and adapt to new invoice formats. Furthermore, every user and system action is recorded for compliance and traceability via audit logging. Figure 2 below demonstrates how the previously discussed features would be used within the system.

Figure 2

Solution Process Flow



The system is uniquely tailored to ADS's infrastructure to include Okta SSO and Alteryx integration while remaining more affordable and customizable than commercial tools, such as Tipalti or SAP Concur. AI² is built with CMMC and FedRAMP compliance in mind in that it is engineered for secure deployment in government-regulated environments. More comparisons can be seen under Table 1 with how well AI² performs against competitors in the market.

Table 1

Competition Matrix

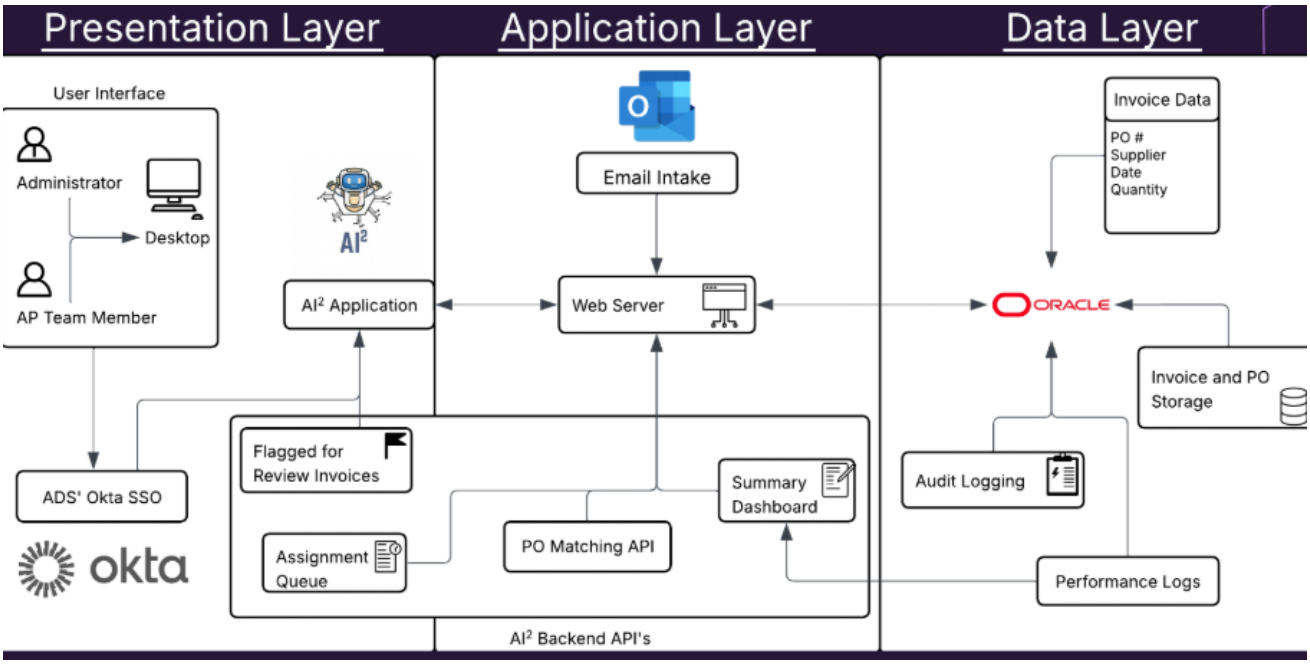
	AI²	Tipalti	SAP Concur	QuickBooks	Manual Processing
Volume handling	🟢	🟢	🟢	🟡	🔴
Error reduction	🟢	🟡	🟢	🟡	🔴
Less human interaction	🟢	🟢	🟢	🟡	🔴
Timeliness	🟢	🟢	🟢	🟡	🔴
AI PO matching	🟢	🟢	🟢	🔴	🔴
AI confidence scoring	🟢	🟢	🟢	🔴	🔴
Continuous learning from past corrections	🟢	🟢	🟢	🔴	🔴
Quick implementation	🟢	🟢	🔴	🟢	🟢
Tailored for client	🟢	🟢	🟢	🔴	🟢
Long-tail vendor handling	🟢	🟢	🟢	🔴	🟢
Easily customizable	🟢	🟢	🔴	🔴	🔴
Adaptable	🟢	🟢	🟢	🔴	🟢

2.3 Major Components (Hardware/Software)

The system's hardware environment is hosted in AWS GovCloud for security and government compliance with end users being able to access it through standard office workstations or laptops. AI²'s software architecture is web-based and built on a secure backend using REST APIs, enabling modular integration with external systems. Key integrations include Outlook invoice capture via Microsoft Graph API, Oracle NetSuite via Alteryx for data exchange and auditability, and document-scanning services such as AWS Textract for PDFs and Aperture for Spreadsheets. Additionally, the AI component itself of AI² consists of a confidence scoring engine and retraining pipeline that together support adaptive learning from user feedback. In turn, the structured invoice data, logs, and metadata are stored in a PostgreSQL database hosted on AWS RDS. Figure 3 below shows a three-tier layout of how these key components would interact.

Figure 3

Major Functional Components Diagram



3 Identification of Case Study

ADS is a government contracting firm managing procurement and supplier payments, making accurate and timely invoice processing essential to its operations. Currently, the organization is facing several issues to include a fully manual invoice processing, a heavy workload for the Accounts Payable staff, and frequent delays to supplier payments caused by the time-consuming nature of the process. They especially experience a significant operational strain due to the manual intake of invoices, PO matching, and exception handling, which make them the ideal primary users for evaluating the prototype and providing iterative feedback. AI² is designed to directly address these problems by automating repetitive work, including the intake, sorting, and matching of invoices, providing dashboards for monitoring invoice status, and

maintaining full audit traceability for compliance review to ensure that ADS can meet both internal and regulatory requirements.

A typical AP representative at ADS starts their day off with opening the Outlook GCC High email inbox that is filled with thousands of unread emails, each containing invoices in different formats. These formats could range anywhere from being in the body of an email to a PDF file to even a spreadsheet. With AI², the representative would only need to log into their dashboard and see that all the new invoices of the day have already been captured, parsed, and organized by supplier. Additionally, they would see that the system has already matched most of their invoices to their purchase orders with high confidence, leaving only a small set of low-confidence invoices that the representative would need to review. After correcting a few mismatches, the representative can watch how, over time, the system learns from those adjustments by the representative and updates future predictions. In turn, what once took hours of manual searching, sorting, and data entry has now evolved to a much improved, more efficient process of targeted review and approval that allows the representative to focus on exceptions rather than routine processing.

Despite AI²'s focus being on ADS, this system can be used in other organizations. Similar challenges exist among other mid-to-large government contractors, logistics companies, and manufacturers that are struggling with inefficiencies in high-volume Accounts Payable workflows. AI²'s modular, integration-ready architecture allows adaptation to other organizations with similar high-volume AP processes and offers a scalable solution for any environment where manual invoice processing is creating bottlenecks, delays, or compliance risks.

Moving forward, AI² could expand beyond invoice-to-PO matching to include integration with additional ERP systems. Future versions could also support mobile approvals and cross-departmental workflow automation. Furthermore, as the system matures, AI² could serve not just ADS but also a broader customer base across government contracting and large-scale financial operations.

4 Glossary

- **AI² (Artificially Intelligent Invoices):** The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.
- **Alteryx:** A data automation and analytics tool that enables secure API connections and data workflows between applications.
- **AP (Accounts Payable):** The amount a business owes for goods and services purchased on credit; typically due at intervals of 30, 45, 60, or 90 days depending on repayment terms.
- **APEX:** A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.
- **AWS GovCloud:** A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.
- **Atlantic Diving Supply (ADS):** An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.
- **CMMC (Cybersecurity Maturity Model Certification):** A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.
- **ERP (Enterprise Resource Planning System):** An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

- **FedRAMP (Federal Risk and Authorization Management Program):** A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.
- **Graph API:** A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.
- **Invoice:** A document listing goods or services provided, including a statement of the amount due; a bill.
- **Oracle:** The system of record used by ADS for enterprise data and financial management.
- **PO (Purchase Order):** The official order list ADS sends to suppliers detailing the items or services requested.
- **Salesforce:** A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.
- **Supplier:** A person or organization that provides goods or services needed by another organization.
- **Vendor:** A person or company offering goods or services for sale.

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ADS Pitch Document Sample

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